REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
KINDOROKO, ONE WORLD AND BISHOP ALPHA SECONDARY SCHOOLS
FORM FOUR SECOND PRE–MOCK EXAMINATIONS

033/2A **BIOLOGY 2A**

(ACTUAL PRACTICAL A)

(For Both Private and School Candidates)

Time: 2:30 Hours **Year: 2025**

Instructions

1. This paper consists of **two (2)** questions. Answer **all** questions.
2. Each question carries **25** marks.
3. Except for diagrams that must be in pencil, all writing must be in blue or black ink.
4. Cellular phones and any unauthorized materials are **not** allowed in the assessment room.
5. Write your **Name in Full** on every page of your answer booklet(s).

BIO JOINT PRE MOCK 2 2025

APR2025

1. You have been provided with specimen **D**, bench reagents and different apparatii. Prepare the solution from specimen **D** using the apparatii provided and label it as **solution D.** Use the reagents provided **only** to test for the common nutrients present in the solution you have prepared and then answer the questions that follow.

   (a) Present your results as shown in the table below.

FOOD TESTED	PROCEDURE	OBSERVATION	INFERENCE

   (b) Stating the apparatus used, outline the steps you have followed when preparing solution **D**.

   (c) (i) Which of the nutrients identified in solution **D** is used to make hormones and antibodies?

   (ii) Name any other two plant sources of the nutrient described in c(i) above.

   (d) (i) Why was it necessary to use hydrochloric acid when testing for some sugars?

   (ii) Give the reason for the series of color changes which is usually observed during the test for reducing and non-reducing sugars.

   (e) (i) What are the end products when each of the nutrients identified in solution **D** is fully digested along the alimentary canal?

   (ii) Explain what happens to each of the products named in e(i) above in the human body when they are found in excess amounts.

   (f) Ileum is the special part for maximum absorption of the final products of digestion. Outline three adaptation of this part for such function.

2. You have been provided with specimens **S, V, T, K** and **U.** Study them carefully and answer the following questions.

   (a) (i) Identify each of the specimens provided by its common name.

   (ii) Classify each of the specimens **T, K** and **U** to class level.

   (b) In which three ways do farmers benefit from the presence of specimen **T** in their farms?

   (c) Describe how the shape of specimen **K** enables it to locomote in its habitat.

   (d) Outline three observable features possessed by specimen **U** that are distinctive for the class in which the specimen belongs.

   (e) (i) Identify the appearance of the leaves of specimen **V** which introduces it to the division it belongs.

   (ii) Closely observe specimen **S** and then identify the floral arrangements which are typical characteristics of the class it belongs.

   (f) Carefully remove ***only the petals*** from specimen **S** using a scalpel and then dissect the remaining specimen longitudinally into two equal halves. Observe one of the halves with the dissected side facing upwards. Draw its diagram and label any four parts you have observed.

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